EXPERIMENT 4: Set Up a One-Node Hadoop Cluster

1. Aim

To install and configure Hadoop in pseudo-distributed mode and establish a single-node Hadoop cluster.

2. Procedure

1. Install Ubuntu Linux and open the Terminal.
2. Update the package repository using sudo apt update.
3. Install a compatible Java Development Kit and verify it using java -version.
4. Install the OpenSSH server package.
5. Generate an SSH key pair and configure passwordless SSH access to localhost.
6. Download a Hadoop binary distribution from the Apache Hadoop website and extract it.
7. Set HADOOP_HOME, JAVA_HOME and the Hadoop bin/sbin directories in the shell environment.
8. Reload the environment configuration and verify Hadoop using hadoop version.
9. Edit etc/hadoop/core-site.xml to set the default filesystem to hdfs://localhost:9000.
10. Edit etc/hadoop/hdfs-site.xml and set dfs.replication to 1 for a single-node setup.
11. Configure mapred-site.xml to use the YARN framework.
12. Configure yarn-site.xml to enable the MapReduce shuffle service.
13. Format the NameNode once using hdfs namenode -format.
14. Start HDFS using start-dfs.sh and start YARN using start-yarn.sh.
15. Run jps and verify that the NameNode, DataNode, SecondaryNameNode, ResourceManager and NodeManager processes are running.

3. Program

Install packages and configure SSH:

```
sudo apt update
sudo apt install openjdk-11-jdk openssh-server -y
ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

Example environment variables (add to ~/.bashrc):

```
export HADOOP_HOME=$HOME/hadoop-3.3.6
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

source ~/.bashrc
```

core-site.xml:

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://localhost:9000</value>
  </property>
</configuration>
```

hdfs-site.xml:

```
<configuration>
  <property>
    <name>dfs.replication</name>
    <value>1</value>
  </property>
</configuration>
```

mapred-site.xml:

```
<configuration>
  <property>
    <name>mapreduce.framework.name</name>
    <value>yarn</value>
  </property>
</configuration>
```

yarn-site.xml:

```
<configuration>
  <property>
    <name>yarn.nodemanager.aux-services</name>
    <value>mapreduce_shuffle</value>
  </property>
</configuration>
```

Start and verify:

```
hdfs namenode -format
start-dfs.sh
start-yarn.sh
jps
```

4. Output of Program

```
Sample jps output:
NameNode
DataNode
SecondaryNameNode
ResourceManager
NodeManager
Jps

NameNode web interface: http://localhost:9870
ResourceManager web interface: http://localhost:8088
```

Note: The Java path and Hadoop version may differ on the lab computer. Update JAVA_HOME and configuration paths accordingly. Format the NameNode only when initializing a fresh cluster.

5. Result

Thus, Hadoop was installed and configured in pseudo-distributed mode, and a single-node Hadoop cluster was established.

EXPERIMENT 4: Set Up a One-Node Hadoop Cluster

1. Aim

To install and configure Apache Hadoop in pseudo-distributed mode on a Linux system and establish a single-node Hadoop cluster for processing distributed data using HDFS and YARN.

2. Procedure

1. Start the Linux virtual machine and log in using the configured user account. 2. Open the terminal and update the package repository using sudo apt update. 3. Install a compatible Java Development Kit because Hadoop requires Java to execute its services. 4. Verify the Java installation using java -version. 5. Install the OpenSSH server package using the appropriate package manager. 6. Generate an SSH key pair using ssh-keygen and configure passwordless SSH access to localhost. 7. Download a compatible Apache Hadoop binary distribution from the official Apache Hadoop website. 8. Extract the downloaded archive into the home directory and identify the Hadoop installation path. 9. Configure the JAVA_HOME and HADOOP_HOME environment variables in the shell configuration file. 10. Add Hadoop's bin and sbin directories to the system PATH. 11. Reload the shell configuration using source ~/.bashrc and verify the Hadoop installation using hadoop version. 12. Open the Hadoop configuration directory and edit core-site.xml to specify the default HDFS filesystem URI. 13. Edit hdfs-site.xml and configure the replication factor as 1 because only one DataNode is being used. 14. Configure mapred-site.xml to use YARN as the execution framework. 15. Edit yarn-site.xml and enable the MapReduce shuffle service. 16. Save all configuration files and verify that their XML structure is correct. 17. Format the NameNode using hdfs namenode -format to initialize the HDFS metadata. 18. Start the HDFS services using start-dfs.sh. 19. Start the YARN services using start-yarn.sh. 20. Execute the jps command to verify that all required Hadoop Java processes are running. 21. Open the NameNode web interface using http://localhost:9870 to inspect the HDFS cluster. 22. Open the ResourceManager web interface using http://localhost:8088 to inspect the YARN resource management service.

4. Output of Program

Expected Output of jps: NameNode DataNode SecondaryNameNode ResourceManager NodeManager Jps Web Interfaces: - NameNode: http://localhost:9870 - ResourceManager: http://localhost:8088 Explanation: The NameNode manages the metadata of files stored in HDFS, while the DataNode stores the actual data blocks. YARN manages cluster resources and executes processing tasks. Since all services are running on the same machine, this configuration is known as a pseudo-distributed Hadoop cluster.

5. Result

Thus, Apache Hadoop was successfully installed and configured in pseudo-distributed mode. The NameNode, DataNode, SecondaryNameNode, ResourceManager and NodeManager services were started and verified, establishing a functional single-node Hadoop cluster.